

THE HR DIAGRAM

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Late in the nineteenth century, astronomers had tools that revealed a great deal about stars. They were able to take spectral images of stars which revealed their composition and they could determine the distances of the closer stars through their angular displacements (parallax) as the Earth orbited the Sun. In 1911, the Danish astronomer Ejnar Hertzsprung (1873-1967) plotted a color- magnitude diagram, discovering the distinction between main sequence and giant stars while studying the Hyades cluster. Two years later, the American astronomer Henry Norris Russell (1877-1957), using similar techniques, was able independently to distinguish between giant and main sequence stars.

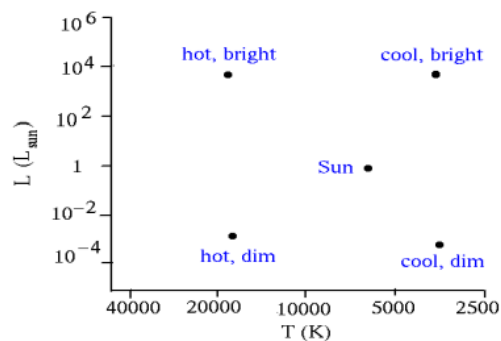


Figure 1. Basic HR Diagram Layout

Here, the x-axis is in Kelvin, but sometimes stellar color or spectral class is used. The y-axis is in units of the Sun's luminosity, but absolute magnitude is employed very often.

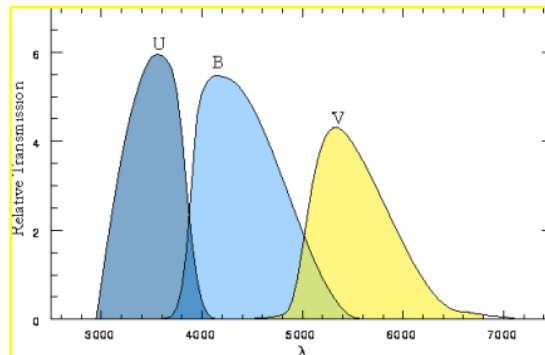
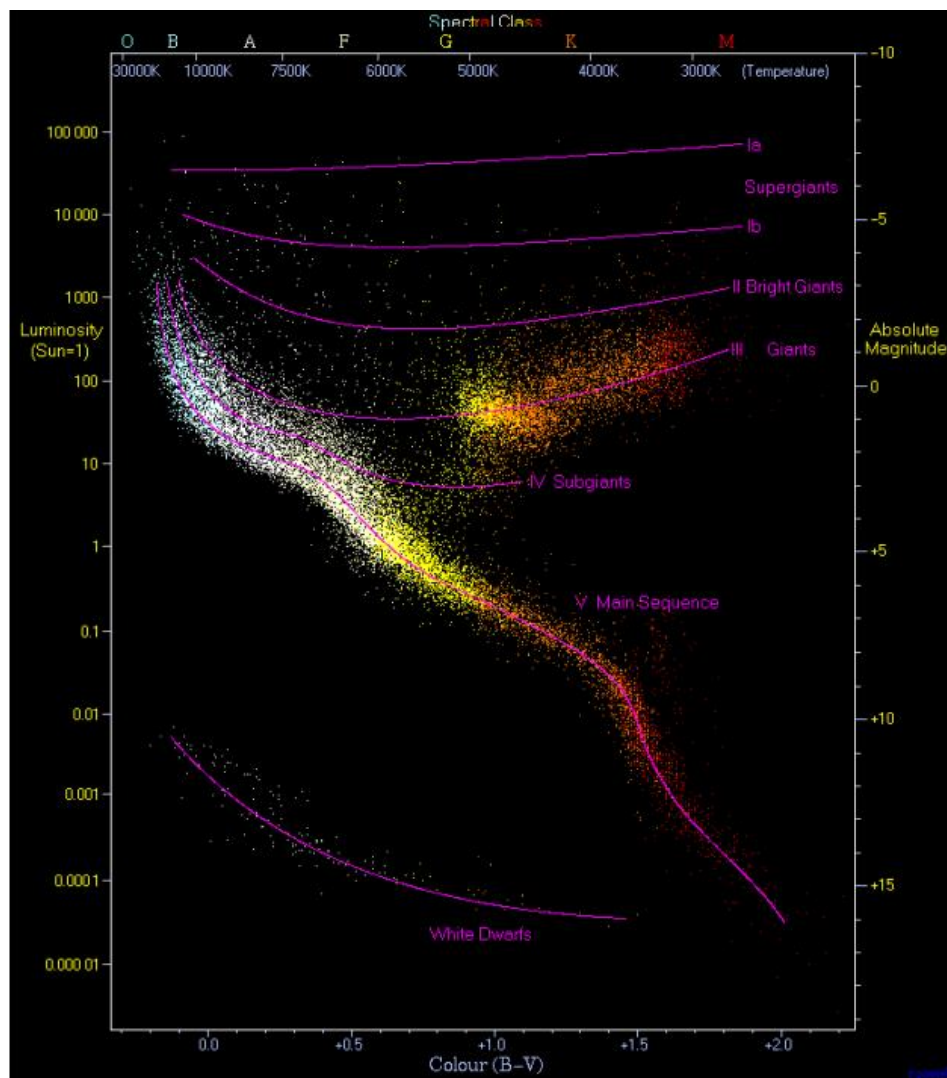


Figure 2. UVB Bandpass, Angstroms

Figure 2 illustrates the wavelength response of standard U (ultraviolet), B (blue) and V (visual) filters. Magnitude measures through such filters are used to create a color index (CI) for a star

Spectral classes in HR diagrams are described in Table 1. Each class is divided into ten parts, 0-9, also in order of decreasing temperatures. For example, the Sun is in the G2 spectral class.

The Harvard Spectral Sequence			
Spectral Class	Distinguishing Characteristics	Temperature (K)	Example
O	Ionized He and metals; weak H	28,000 - 60,000	zeta Orionis
B	Neutral He, ionized metals, stronger H	10,000 - 28,000	Rigel, Spica
A	Balmer H dominant, singly-ionized metals	7500 - 10,000	Sirius, Deneb
F	H weaker, neutral and singly-ionized metals	6000 - 7500	Procyon, Canopus
G	Singly ionized Ca, H weaker, neutral metals	5000 - 6000	Sun, Capella
K	Neutral metals, molecular bands begin to appear	3500 - 5000	Aldebaran, Arcturus
M	Ti oxide molecular lines; neutral metals	< 3500	Antares, Betelgeuse



The ordinary hydrogen- burning dwarf stars like the Sun are found in a band running from top-left to bottom-right called the Main Sequence. Giant stars form their own clump on the upper-right side of the diagram. Above them lie the much rarer bright giants and supergiants. At the lower-left is the band of white dwarfs - these are the dead cores of old stars which have no internal energy source and, over billions of years, they slowly cool down towards the bottom-right of the diagram.

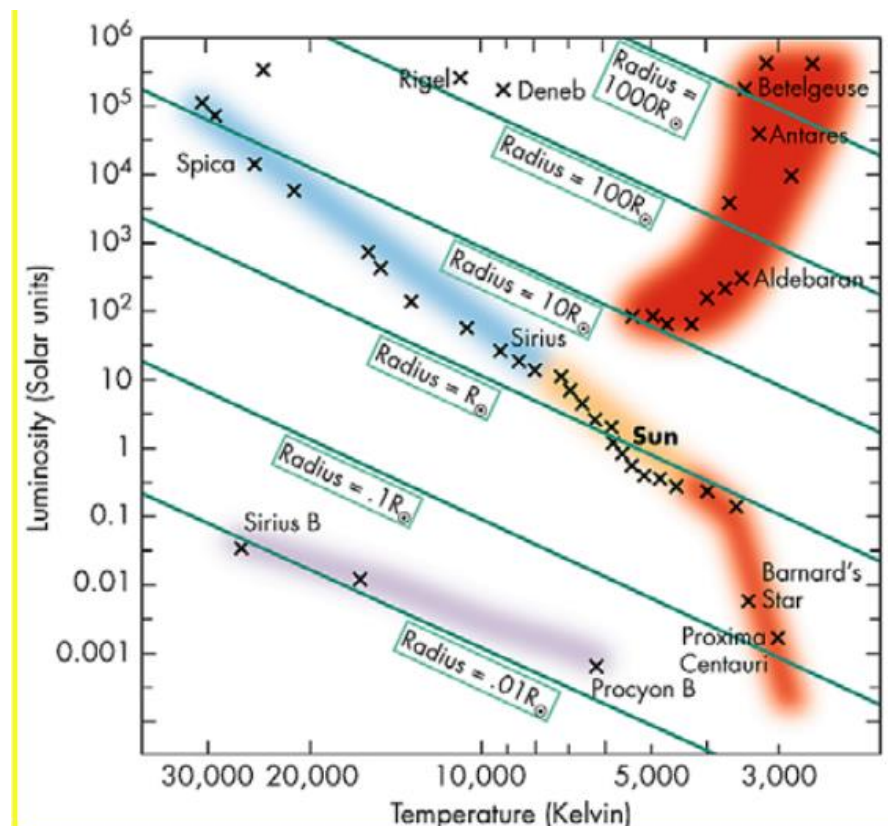
I **Supergiants**: Stars that are very massive and very luminous near the end of their lives. They are subclassified as Ia or Ib, with Ia representing the brightest of these stars. These stars are very rare - one in a million stars is a supergiant. The nearest supergiant star is Canopus (F0Ib) 310 light years away. Some other examples are Betelgeuse (M2Ib), Antares (M1Ib) and Rigel (B8Ia).

II **Bright Giants** Stars which have a luminosity between the giant and supergiant stars. Some examples are Sargas (F1II) and Alpheratz (K3II).

III **Normal Giants** These are mainly low-mass stars at the end of their lives that have swelled to become a giant star. This category also includes some high mass stars evolving on their way to supergiant status. Some examples are Arcturus (K2III), Hadar (B1III) and Aldebaran (K5III).

IV **Subgiants** Stars which have begun evolving to giant or supergiant status. Some examples are Alnilam (B7IV) and Mirzam (G0IV).

V **Dwarfs** (Main Sequence and White Dwarfs) All normal hydrogen-burning stars. Stars spend most of their lives in this category before evolving up the scale. Class O and B stars in this category are actually very bright and luminous and are generally brighter than most Giant stars. Some examples are the Sun (G2V), Sirius (A1V), and Vega (A0V).



The slope of the line occurs because luminosity varies as T^4 . Note that while Rigel and Betelgeuse have about the same luminosity, Rigel is about 3x hotter while Betelgeuse is about 9x larger.

Stellar Evolution describes the lifecycle of stars, divided into pre-main sequence, main sequence, and post-main sequence stages:

1. Pre-Main Sequence:

Initial Collapse: A cloud of gas collapses under gravity, heating up to form a large, cool protostar. Despite its low temperature, its large surface area results in high luminosity (100× its main-sequence luminosity).

Gravitational Contraction: The protostar radiates energy and contracts rapidly, increasing temperature while overall luminosity decreases, causing a vertical drop on the H-R diagram.

Hydrogen Fusion Ignition: Once the core reaches 10 million K, nuclear fusion of hydrogen into helium begins. Initial radiation pressure slows contraction and slightly increases surface temperature and luminosity.

Hydrostatic Equilibrium: Balance between gravitational pull and outward pressure from fusion stabilizes the star, which settles on the **Zero Age Main Sequence (ZAMS)**.

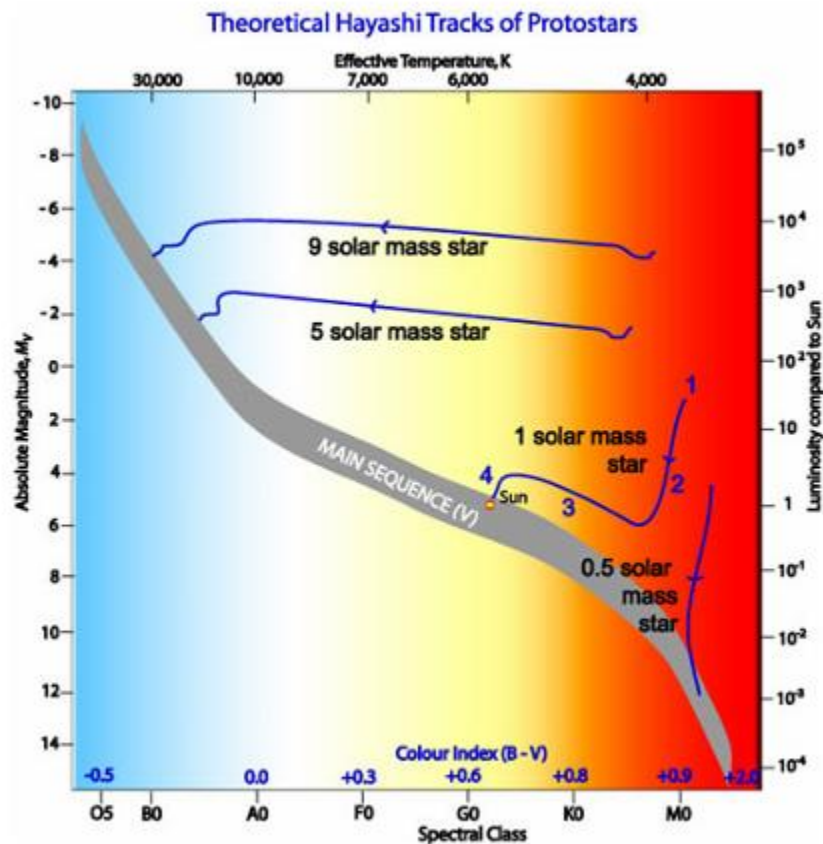


Figure 6. Protostars Entering The Main Sequence

2. Post-Main Sequence:

Hydrogen Depletion: Stars exhaust hydrogen in their cores, leaving the main sequence. Their surface temperature decreases, size increases, and they evolve into red giants or supergiants.

Mass-Dependent Fate:

- **High-Mass Stars:** Become supergiants, potentially ending as black holes or neutron stars.
- **Medium-Mass Stars (e.g., the Sun):** Expand to red giants, lose outer layers, and become white dwarfs.
- **Small Stars:** End as dim white dwarfs after expanding minimally.

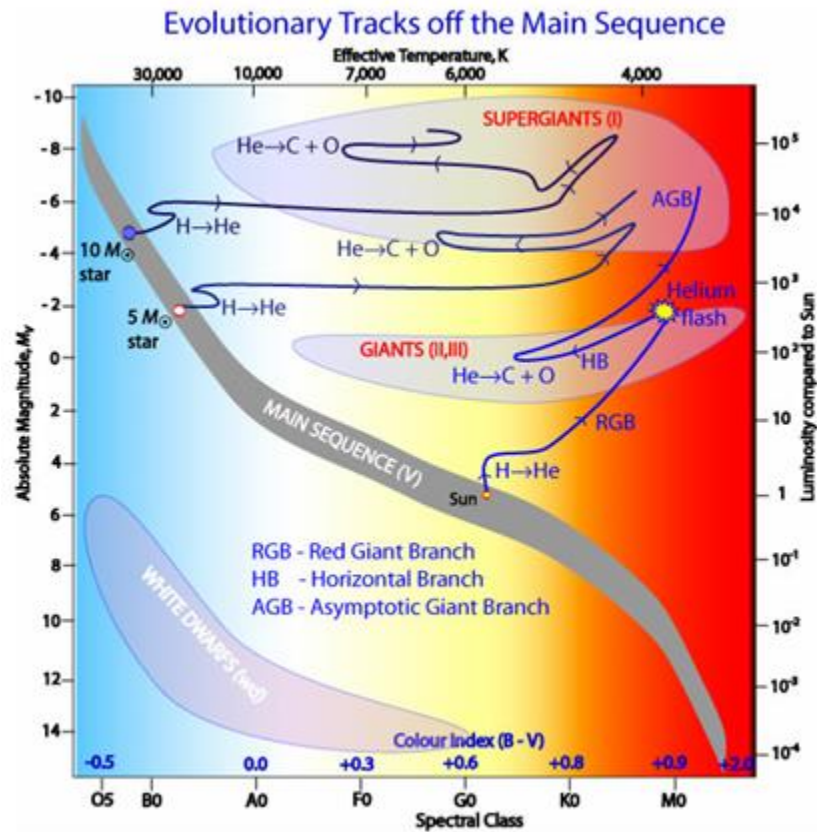
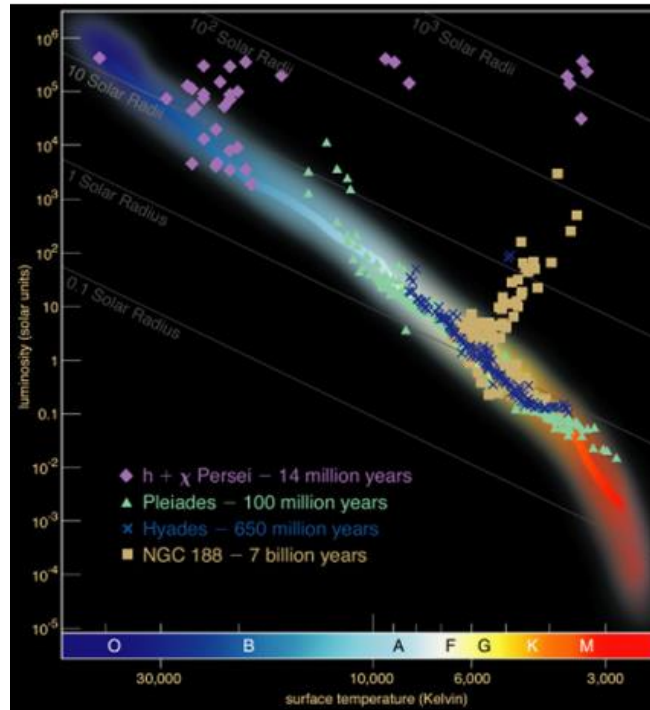


Figure 7. Stars Leaving the Main Sequence

3. Evolution in Star Clusters:

Turnoff Point: The location where stars begin leaving the main sequence indicates cluster age.

- Higher turnoff points (massive stars leaving the main sequence) mean younger clusters.
- Example: Hyades (~650 million years old) has a turnoff at A3 spectral class.
 - Globular clusters, much older than open clusters, exhibit no massive main-sequence stars, with turnoff points in lower-temperature classes like G.



Star clusters are formed from giant molecular clouds and dust. As a result, the stars in a cluster are the same age and about the same distance from Earth, but they develop in a range of masses. We can assess the age of each cluster by its turnoff point, the place where its stars begin to leave the main sequence. The higher (more luminous) the turnoff point on the main sequence, the younger the cluster. This is because more luminous stars on the main sequence, being more massive, have shorter lifetimes. The Pleiades cluster's turnoff point is at about B6, for a cluster age of ~100 million years. Globular clusters are much older than open clusters. When plotted on an HR diagram, they typically have no remaining O, B, A, F class main sequence stars, and their turnoff occurs in the G class. This implies that ages of some globular clusters are close to that of the universe.

In the century since the bases of HR diagrams were first developed, they have become invaluable in the study of stars. For astronomers, a star's location in an HR diagram provides the same depth of insight as for chemists examining an element in the periodic table. It relates hydrogen fusing stars' temperatures, colors and spectral classes with their luminosities, thereby providing insights into their evolution. A star's position in the diagram can indicate its composition, surface temperature, intrinsic brightness, size, mass, lifetime, and evolutionary status. That's a lot of information from a diagram. It's been invaluable in determining the age of various star clusters. Apparent magnitudes can be used if absolute magnitudes are not available, since all stars in a cluster are roughly equally distant.

Geophysical tests for habitability in ice-covered ocean worlds

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****1. Objective**:** This study focuses on using geophysical measurements to understand the internal structure and habitability of icy ocean worlds, such as Europa, Enceladus, Titan, Ganymede, and Callisto. Understanding these factors is essential to finding environments that might support life beyond Earth.

****2. Parameters Measured**:** The research looks at important geophysical factors like density, temperature, sound speed, and electrical conductivity. These help describe the physical and chemical conditions of these ocean worlds and their potential for supporting life.

****3. Thermodynamic Constraints****: To create models of the internal structures of these celestial bodies, the authors used one-dimensional (1-D) calculations based on available thermodynamic data related to aqueous solutions like magnesium sulfate (MgSO₄) and sodium chloride (NaCl), along with different types of water ices and silicate materials. However, they note that the current thermodynamic data has limitations, particularly in pressure, temperature, and composition, making it hard to understand the salinity conditions for both MgSO₄ and NaCl. This lack of data creates gaps in knowledge about mixtures of sodium, magnesium, chlorine, sulfate, and ammonia.

****4. Specific Findings****:

- ****Europa****: The study suggests that Europa's ocean may have originally been hot and rich in reducing chemicals, which could support life. Over time, it likely became oxidized due to radiolytic materials from its surface.

- ****Enceladus****: Enceladus may contain hydrated minerals or highly porous anhydrous minerals, similar to Earth's upper mantle, which shows how water and rock interact and could support life.

- ****Titan****: Titan likely has a dense ocean that does not need to be saline. The study mentions that it might not have high-pressure ice formations, which impacts our understanding of its ocean characteristics.

- ****Ganymede****: Ganymede is thought to have the deepest siliceous interior among the known ocean worlds, possibly containing different forms of high-pressure ice that could float if the ocean is salty enough.

- ****Callisto****: The composition of Callisto's ocean is still unclear because of insufficient data. The authors state that it might have an ocean near its eutectic composition and may lack high-pressure ices, but this is uncertain due to unknown factors about its structure.

****5. Geophysical Signatures****: The study highlights how geophysical signatures can help identify conditions that might support life. These signs are essential for future exploration of icy bodies and improving our understanding of their potential habitability.

****6. Importance****: This research shows the need for better thermodynamic data and offers important tools for planning future geophysical investigations of icy ocean worlds. By enhancing our knowledge of these unique environments, the study adds to the search for life beyond Earth and sheds light on the nature of ocean worlds in our solar system.

In conclusion, this examination illustrates the connection between geophysical factors, internal structure, and habitability in icy ocean worlds while pointing out the critical need for more data to fully understand these fascinating celestial bodies.